Classtes. Digital Logic design. Time : 30 min. Marks: 30

Answer any 2(Two) of the following questions:

1. Write the truth table, min. Boolean expression of a full adder with circuit diagram. Show how 6-bit parallel adder can be constructed using 2-bit parallel adder. 15
2. Design a 4-input multiplexer. Implement the following Boolean expression using 4:1 multiplexer block. 15

$$\bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}CD + A\bar{B}CD + \bar{A}BCD + AB\bar{C}D + ABC\bar{D} + AB\bar{C}\bar{D}$$

3. a) Design a decimal to BCD encoder. b) Implement a 4:16 decoder using 3:8 decoder blocks. 15

Scanned with CamScanner

Tama shil

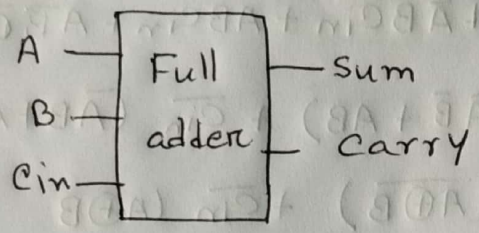
ID: 342

T-2

Answer to the question NO: 01.

Full adder

Full adder accepts two input bits and an input carry and generate a sum output and output carry.



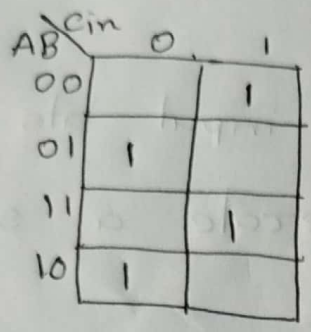
Truth table-

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Scanned with CamScanner

Expression using k-map

for sum -



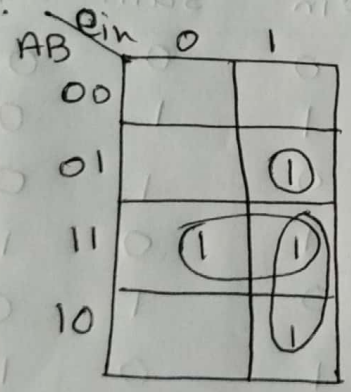
$$Sum = \bar{A}\,\bar{B}C_{in} + \bar{A}B\bar{C_{in}} + ABC_{in} + A\bar{B}\,\bar{C_{in}}$$

$$= C_{in}(\bar{A}\bar{B} + AB) + \bar{C_{in}}(\bar{A}B + A\bar{B})$$

$$= C_{in}(\overline{A \oplus B}) + \bar{C_{in}}(A \oplus B$$

$$= (A \oplus B) \oplus C_{in}$$

for Cout →



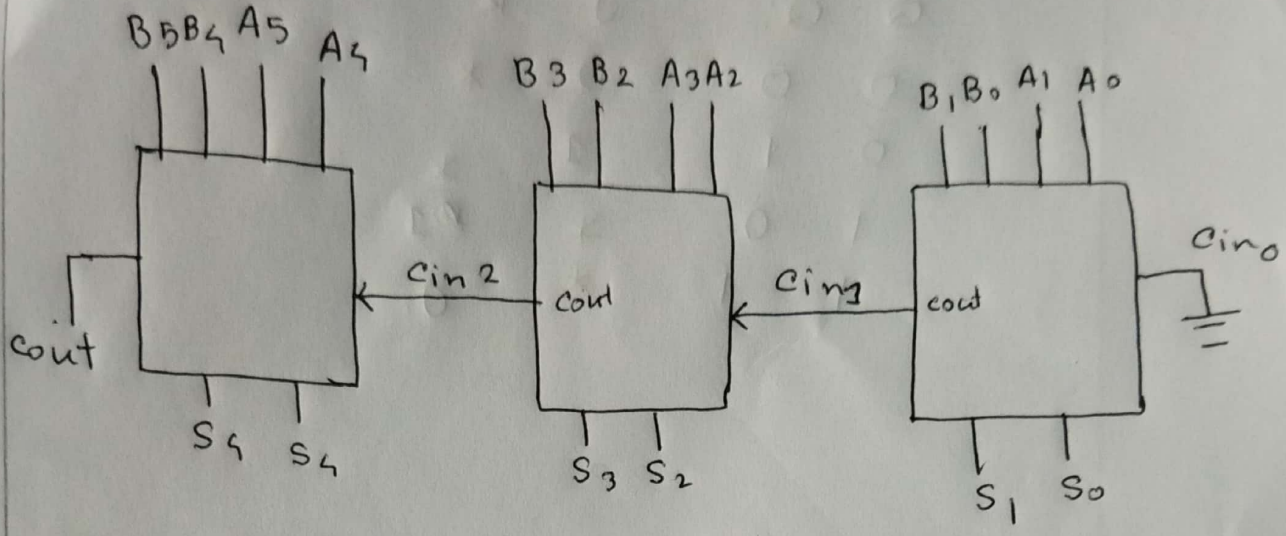
$$C_{out} = AB + \bar{A}BC_{in} + A\bar{B}C_{in}$$

$$= AB + C_{in}(\bar{A}B + A\bar{B})$$

$$= AB + C_{in}(A \oplus B)$$

Scanned with CamScanner

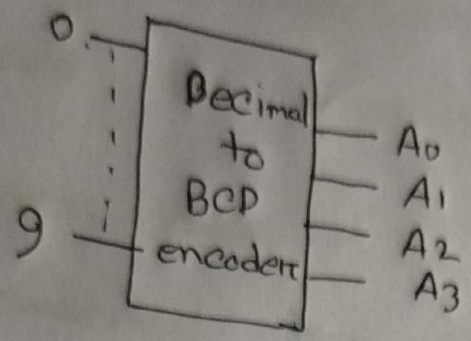
6 bit parallel adder from 2 bit parallel adder:



Scanned with CamScanner

Answers to the question No: 03

a) Decimal to BCD encoder



truth table

input	output			
	A3	A2	A1	A0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1

P Expression

$A_0 = 1+3+5+7+9$

$A_1 = 2+3+6+7$

$A_2 = 4+5+6+7$

$A_3 = 8+9$

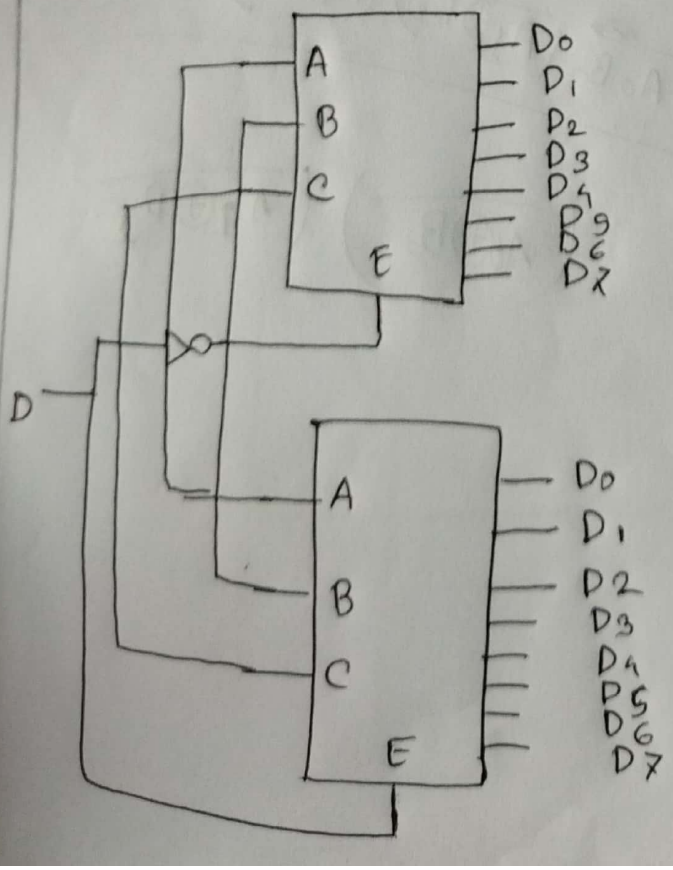
Scanned with CamScanner

b) Implimenting 4:16 decoder using 3:8 decoder-

4:16 decoder circuit can be obtained from two 3:8 decoder circuits.

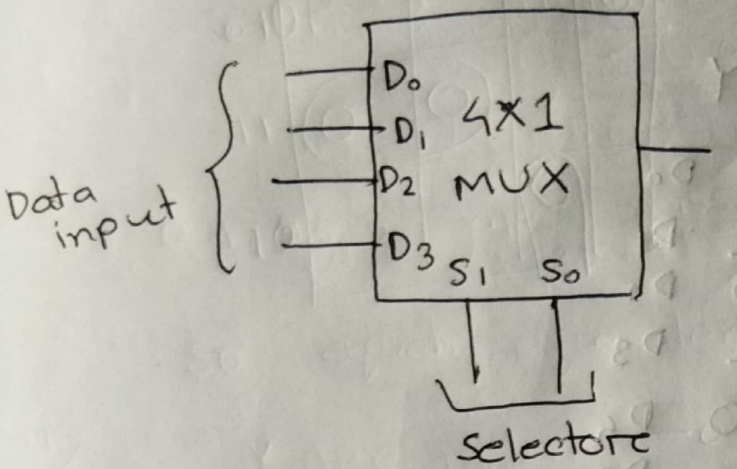
When two 3:8 decoder circuits are combined the enable pin acts as the input for both decoders.

When ~~ea~~ enable pin is high at one 3 to 8 decoder circuits then its low is at another 3 to 8 decoder circuit



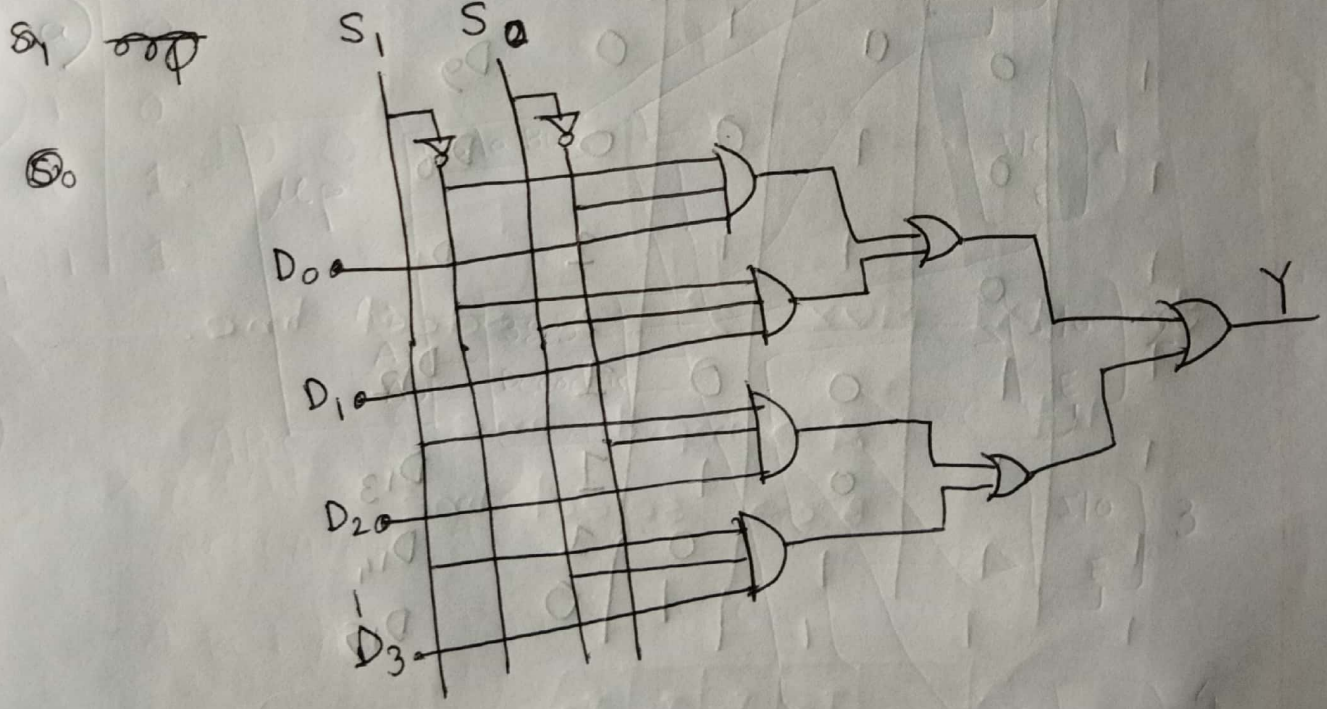
Scanned with CamScanner

Design a 4 input multiplexer



S_1	S_0	Y
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3

$$Y = \bar{S}_0\bar{S}_1D_0 + \bar{S}_1S_0D_1 + S_1\bar{S}_0D_2 + S_1S_2D_3$$



Scanned with CamScanner

$\bar{A}\bar{B}\bar{C}D + A\bar{B}\bar{C}D + A\bar{B}CD + \bar{A}BCD + \bar{A}B\bar{C}D + ABCD + ABC\bar{D}$

A	B	C	D	Y	
0	0	0	0	0	D_0
0	0	0	1	1	D_1
0	0	1	0	0	D_2
0	0	1	1	1	D_3
0	1	0	0	0	D_4
0	1	0	1	0	D_5
0	1	1	0	0	D_6
0	1	1	1	1	D_7
1	0	0	0	0	D_8
1	0	0	1	0	D_9
1	0	1	0	0	D_{10}
1	0	1	1	1	D_{11}
1	1	0	0	1	D_{12}
1	1	0	1	1	D_{13}
1	1	1	0	1	D_{14}
1	1	1	1	0	D_{15}

4X1 MUX → 2^2 → 2 select lines

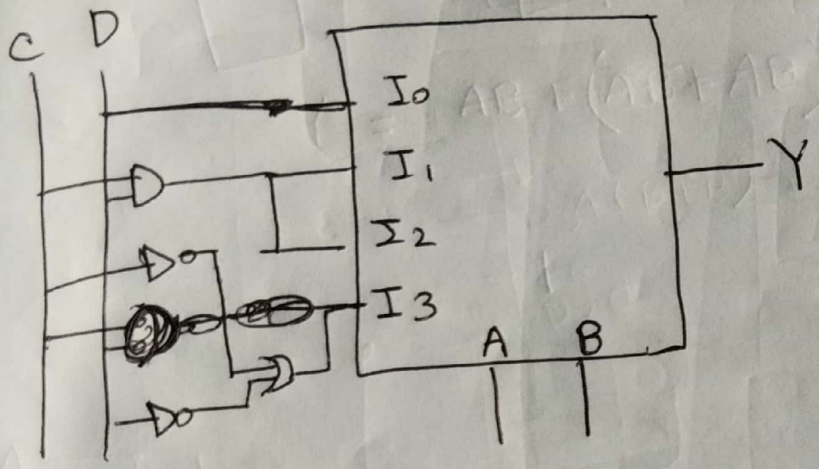
here → select lines → A, B

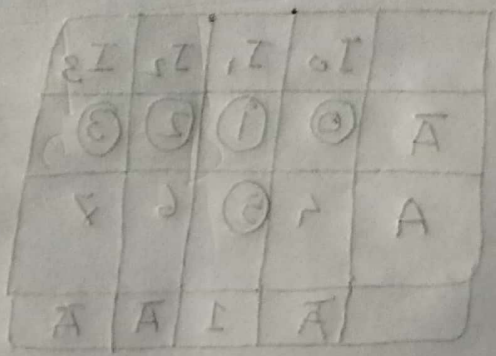
input " → C, D

Scanned with CamScanner

	I_0	I_1	I_2	I_3
$\bar{C}\bar{D}$	0	4	8	(12)
$\bar{C}D$	(1)	5	9	(13)
$C\bar{D}$	2	6	10	(14)
CD	(3)	(7)	(11)	15
	D	CD	CD	$C+D$

$$\bar{C}\bar{D}+\bar{C}D+C\bar{D}$$
$$= \bar{C}(\bar{D}+D)+C\bar{D}$$
$$= \bar{C}+C\bar{D}$$
$$= \bar{C}+\bar{C}\bar{D}+C\bar{D}$$
$$= \bar{C}+\bar{D}$$





Scanned with CamScanner